ESXi

SAN Host

NetApp
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ESXi

适用于采用 ONTAP 的 ESXi 7.x 的 NVMe-oF 主机配置

可支持性

ONTAP 9.7 或更高版本支持基于光纤通道的 NVMe （ NVMe/FC ）。

功能

- ESXi 启动程序主机可以通过相同的适配器端口同时运行 NVMe/FC 和 FCP 流量。请参见 "Hardware Universe" 有关支持的 FC 适配器和控制器的列表。请参见 "NetApp 互操作性表" 有关支持的最新配置和版本列表。
- 从 ONTAP 9.9.1 P3 开始， ESXi 7.0 Update 3 支持 NVMe/FC 功能。
- 对于 ESXi 7.0 及更高版本， HPP （高性能插件）是 NVMe 设备的默认插件。

已知限制

不支持以下配置：

- RDM 映射
- VVOL

启用 NVMe/FC

1. 检查 ESXi 主机 NQN 字符串，并验证它是否与 ONTAP 阵列上相应子系统的主机 NQN 字符串匹配：

```
# esxcli nvme  info get
Host NQN: nqn.2014-08.com.vmware:nvme:nvme-esx

# vserver nvme subsystem host show -vserver vserver_nvme
  Vserver Subsystem             Host NQN
  ------- -------------------- ----------------------------------------
  vserver_nvme ss_vserver_nvme nqn.2014-08.com.vmware:nvme:nvme-esx
```

配置 Broadcom/Emulex

1. 请参见，检查所需驱动程序 / 固件是否支持此配置 "NetApp 互操作性表"。
2. 设置 lpfc 驱动程序参数 `lpfc_enable_FC4_type=3` 以在 `lpfc` 驱动程序中启用 NVMe/FC 支持，然后重新启动主机。

> 从 vSphere 7.0 Update 3 开始， `brcmnvmefc` 驱动程序不再可用。因此， `lpfc` 驱动程序现在包括先前随 `brcmnvmefc` 驱动程序提供的基于光纤通道的 NVMe （ NVMe/FC ）功能。

默认情况下，已为 LPe35000 系列适配器设置 `lpfc_enable_FC4_type=3` 参数。要为 LPe32000 系列和 LPe31000 系列适配器手动设置此命令，必须执行以下命令。

```
# esxcli system module parameters set -m lpfc -p lpfc_enable_fc4_type=3

#esxcli system module parameters list  -m lpfc | grep lpfc_enable_fc4_type
lpfc_enable_fc4_type              int     3      Defines what FC4 types
are supported

#esxcli storage core adapter list
HBA Name  Driver   Link State  UID
Capabilities         Description
--------  -------  ----------  -----------------------------------
-------------------  -----------
vmhba1    lpfc     link-up     fc.200000109b95456f:100000109b95456f
Second Level Lun ID  (0000:86:00.0) Emulex Corporation Emulex LPe36000
Fibre Channel Adapter    FC HBA
vmhba2    lpfc     link-up     fc.200000109b954570:100000109b954570
Second Level Lun ID  (0000:86:00.1) Emulex Corporation Emulex LPe36000
Fibre Channel Adapter    FC HBA
vmhba64   lpfc     link-up     fc.200000109b95456f:100000109b95456f
(0000:86:00.0) Emulex Corporation Emulex LPe36000 Fibre Channel Adapter
NVMe HBA
vmhba65   lpfc     link-up     fc.200000109b954570:100000109b954570
(0000:86:00.1) Emulex Corporation Emulex LPe36000 Fibre Channel Adapter
NVMe HBA
```

配置 **Marvell/QLogic**

1. 请参见，检查所需驱动程序 / 固件是否支持配置 "NetApp 互操作性表"。
2. 设置 `qlnativefc` driver 参数 `ql2xnvmesupport=1` 以在 `qlnativefc` 驱动程序中启用 NVMe/FC 支持，然后重新启动主机。

   ` # esxcfg-module -s 'ql2xnvmesupport=1' qlnativefc`

   默认情况下， QLE 277x 系列适配器会设置 `qlnativefc` driver 参数。要为 QLE 277x 系列适配器手动设置此命令，必须执行以下命令。

   ```
   esxcfg-module -l | grep qlnativefc
   qlnativefc                4     1912
   ```

3. 检查适配器上是否启用了 NVMe：

```
  #esxcli storage core adapter list
HBA Name  Driver      Link State  UID
Capabilities         Description
--------  ----------  ----------  ------------------------------------
-------------------  -----------
 vmhba3    qlnativefc  link-up     fc.20000024ff1817ae:21000024ff1817ae
Second Level Lun ID  (0000:5e:00.0) QLogic Corp QLE2742 Dual Port 32Gb
Fibre Channel to PCIe Adapter    FC Adapter
vmhba4    qlnativefc  link-up     fc.20000024ff1817af:21000024ff1817af
Second Level Lun ID  (0000:5e:00.1) QLogic Corp QLE2742 Dual Port 32Gb
Fibre Channel to PCIe Adapter FC Adapter
vmhba64   qlnativefc  link-up     fc.20000024ff1817ae:21000024ff1817ae
(0000:5e:00.0) QLogic Corp QLE2742 Dual Port 32Gb Fibre Channel to PCIe
Adapter  NVMe FC Adapter
vmhba65   qlnativefc  link-up     fc.20000024ff1817af:21000024ff1817af
(0000:5e:00.1) QLogic Corp QLE2742 Dual Port 32Gb Fibre Channel to PCIe
Adapter  NVMe FC Adapter
```

验证 NVMe/FC

1. 验证 NVMe/FC 适配器是否列在 ESXi 主机上：

```
# esxcli nvme adapter list

Adapter  Adapter Qualified Name           Transport Type  Driver
Associated Devices
-------  -------------------------------  --------------  ----------
------------------
vmhba64  aqn:qlnativefc:21000024ff1817ae  FC              qlnativefc
vmhba65  aqn:qlnativefc:21000024ff1817af  FC              qlnativefc
vmhba66  aqn:lpfc:100000109b579d9c        FC              lpfc
vmhba67  aqn:lpfc:100000109b579d9d        FC              lpfc
```

2. 验证是否已正确创建 NVMe/FC 命名空间：

   以下示例中的 UUID 表示 NVMe/FC 命名空间设备。

```
# esxcfg-mpath -b
uuid.5084e29a6bb24fbca5ba076eda8ecd7e : NVMe Fibre Channel Disk
(uuid.5084e29a6bb24fbca5ba076eda8ecd7e)
   vmhba65:C0:T0:L1 LUN:1 state:active fc Adapter: WWNN:
20:00:34:80:0d:6d:72:69 WWPN: 21:00:34:80:0d:6d:72:69  Target: WWNN:
20:17:00:a0:98:df:e3:d1 WWPN: 20:2f:00:a0:98:df:e3:d1
   vmhba65:C0:T1:L1 LUN:1 state:active fc Adapter: WWNN:
20:00:34:80:0d:6d:72:69 WWPN: 21:00:34:80:0d:6d:72:69  Target: WWNN:
20:17:00:a0:98:df:e3:d1 WWPN: 20:1a:00:a0:98:df:e3:d1
   vmhba64:C0:T0:L1 LUN:1 state:active fc Adapter: WWNN:
20:00:34:80:0d:6d:72:68 WWPN: 21:00:34:80:0d:6d:72:68  Target: WWNN:
20:17:00:a0:98:df:e3:d1 WWPN: 20:18:00:a0:98:df:e3:d1
   vmhba64:C0:T1:L1 LUN:1 state:active fc Adapter: WWNN:
20:00:34:80:0d:6d:72:68 WWPN: 21:00:34:80:0d:6d:72:68  Target: WWNN:
20:17:00:a0:98:df:e3:d1 WWPN: 20:19:00:a0:98:df:e3:d1
```

> 在 ONTAP 9.7 中， NVMe/FC 命名空间的默认块大小为 4k 。此默认大小与 ESXi 不兼容。因此，在为 ESXi 创建命名空间时，必须将命名空间块大小设置为 512b 。您可以使用 `vserver nvme namespace create` 命令执行此操作。

```
vserver nvme namespace create -vserver vs_1 -path /vol/nssvol/namespace1 -size
100g -ostype vmware -block-size 512B
```

请参见 "ONTAP 9 命令手册页" 了解更多详细信息。

3. 验证相应 NVMe/FC 命名空间设备的各个 ANA 路径的状态：

```
esxcli storage hpp path list -d uuid.5084e29a6bb24fbca5ba076eda8ecd7e
fc.200034800d6d7268:210034800d6d7268-
fc.201700a098dfe3d1:201800a098dfe3d1-
uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Runtime Name: vmhba64:C0:T0:L1
   Device: uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Device Display Name: NVMe Fibre Channel Disk
(uuid.5084e29a6bb24fbca5ba076eda8ecd7e)
   Path State: active
   Path Config: {TPG_id=0,TPG_state=AO,RTP_id=0,health=UP}

fc.200034800d6d7269:210034800d6d7269-
fc.201700a098dfe3d1:201a00a098dfe3d1-
uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Runtime Name: vmhba65:C0:T1:L1
   Device: uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Device Display Name: NVMe Fibre Channel Disk
(uuid.5084e29a6bb24fbca5ba076eda8ecd7e)
   Path State: active
   Path Config: {TPG_id=0,TPG_state=AO,RTP_id=0,health=UP}

fc.200034800d6d7269:210034800d6d7269-
fc.201700a098dfe3d1:202f00a098dfe3d1-
uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Runtime Name: vmhba65:C0:T0:L1
   Device: uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Device Display Name: NVMe Fibre Channel Disk
(uuid.5084e29a6bb24fbca5ba076eda8ecd7e)
   Path State: active unoptimized
   Path Config: {TPG_id=0,TPG_state=ANO,RTP_id=0,health=UP}

fc.200034800d6d7268:210034800d6d7268-
fc.201700a098dfe3d1:201900a098dfe3d1-
uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Runtime Name: vmhba64:C0:T1:L1
   Device: uuid.5084e29a6bb24fbca5ba076eda8ecd7e
   Device Display Name: NVMe Fibre Channel Disk
(uuid.5084e29a6bb24fbca5ba076eda8ecd7e)
   Path State: active unoptimized
   Path Config: {TPG_id=0,TPG_state=ANO,RTP_id=0,health=UP}
```

已知问题描述

- 从 ONTAP 9.9.1 P3 开始，可提供 ESXi 7.0 U3 （及更高版本） NVMe/FC 支持。这是由于仅在 ONTAP 9.9.1 P3 起提供的关键 NVMe 中止（由 ESXi 7.0 U3 及更高版本发出）修复程序。请参见相应的 burt 公有

报告，网址为 https://mysupport.netapp.com/site/bugs-online/product/ONTAP/BURT/1420654 了解详细信息。

相关链接

"TR-4597： 采用 ONTAP 的 VMware vSphere"
"NetApp MetroCluster 支持 VMware vSphere 5.x， 6.x 和 7.x （2031038）"
"NetApp ® SnapMirror ® Business Continuity （SM-BC）支持 VMware vSphere 6.x 和 7.x"

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